Princeton University Science Olympiad
Water Quality Exam

Part I: Marine and Estuary Ecology (40 points)

1. The ocean contains ___% sodium chloride and ___% other salts. (1 point)
    a. 5; 3
    b. 3; 1
    c. 5; 0.75
    d. 3; 0.5
    e. 0.75; 3

2. Which of the following are components of the secondary treatment process? (1 point, Circle all that apply)
    a. Sand & Grit removal
    b. Aeration
    c. Disinfection
    d. Nutrient removal

3. Name 3 potential consequences of wastewater pollutants, if not treated or removed (3 points).

4. Name an organism that follows an r-type strategy and sketch a rough survival curve for that organism (3 points).

5. Which type of strategy would you expect to be more successful during primary succession? (1 point)
    a. r strategy
    b. K strategy

6. Explain your answer to (5) and whether or not you would expect the success of the chosen strategy to remain stable as succession progresses. (2 points)

7. Which of the following is the greatest source of viral and bacterial contamination in estuaries? (1 point)
    a. Livestock waste
    b. Combined sewage overflows
    c. Sewage treatment plants
    d. Leaky septic systems

8. Explain the extent to which oysters and clams are affected by pathogens in estuary ecosystems. (2 points)

9. Name 2 types of wastewater that feed into combined sewer systems and explain where CSOs direct wastewater. (1 point)

10. Name an adaptation of an estuary organism (or taxa of estuary organisms) and explain how that adaptation responds to a particular selective pressure. (3 points)

Questions 11-15 are based on the picture below:



11. The common name of the organism pictured above: (1 point)

12. How is this organism thought to have been introduced to the United States? (1 point)

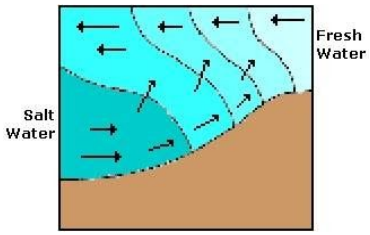
13. The organism pictured above is catadromous. Explain what this means. (2 points)

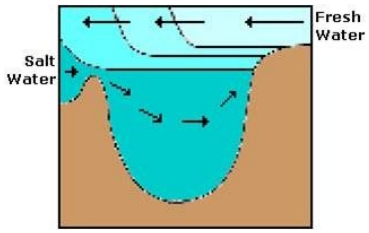
14. Why might this species cause significant ecological disruptions in estuarine areas where it is invasive? Explain briefly in terms of the type of niche it occupies. (2 points)

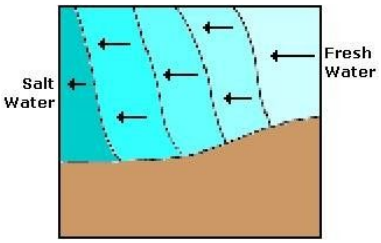
15. How does this species directly interfere with water treatment? (1 point)

16. More than half of the 32 largest cities in the world are constructed on estuaries (True/False). (1 point)

17. Salt-wedge and fjord estuaries are classified differently based on their _____. (1 point)

a. 

b. 

c. 

18. Classify the three estuaries above. (3 points)

19. Name an additional factor that is expected to differ between the estuaries in (18), other than the factor named in (17). (1 point)

20. Indicate which estuary habitat you expect to occur at the highest elevation relative to sea level: (1 point)
    a. Salt flat
    b. Intertidal mudflat
    c. Salt marsh
    d. Mangrove zone

21. The pH of surface ocean waters has decreased by 0.1 on average since the beginning of the industrial revolution. What percentage change in acidity does this represent, to the nearest tenth? (1 point)

22. Explain what effect, if any, dissolved oxygen levels around 5-15 PPM would be expected to have on organisms in surface waters. (2 points)

23. [TIEBREAKER] Describe the purpose of the National Estuarine Research Reserve System, including two of the main activities it facilitates. (3 points)

24. Name the largest reservoir of each element: (2 points)
    a. Carbon:
    b. Nitrogen:
    c. Phosphorus:
    d. Oxygen:

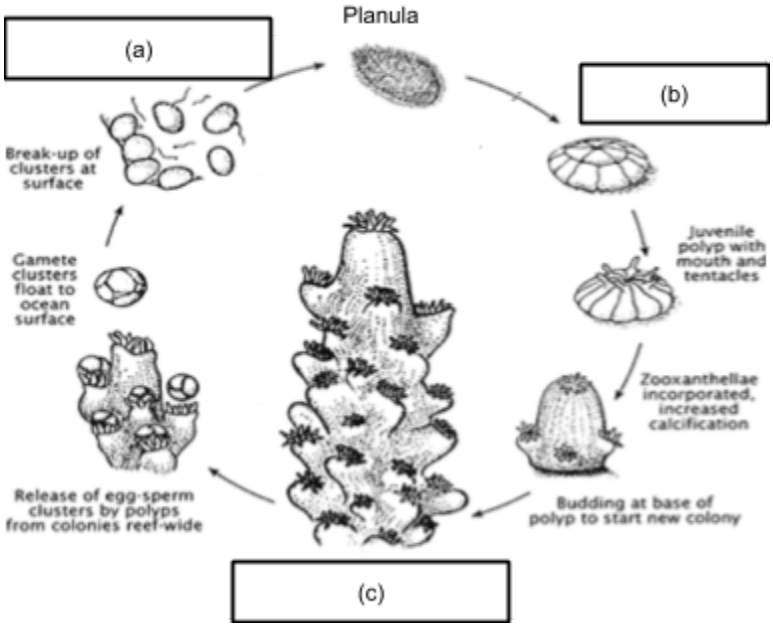
25. Why does ice float? (1 point)

26. What two types of microorganisms are typically present in anaerobic sludge digesters? (2 points)

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Part II (40 points)

1. Complete the below diagram of the lifecycle of a coral polyp: (3 points)



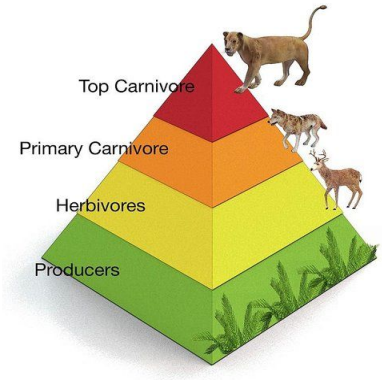
2. Simply explain (i.e., in a sentence or two) what ocean acidification is and how it affects coral reefs. (2 points)

3. What is wrong with the coral in the following image? What is it missing? (2 points)



4. What is one way overfishing can damage a reef ecosystem? (2 points)

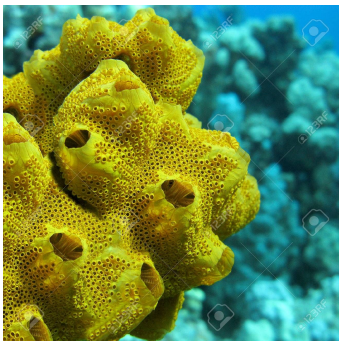
5. The following is an example of a typical trophic pyramid from an arbitrary ecosystem. Why does it make sense to represent these relations with a pyramid? In other words, what does a trophic pyramid represent? (2 points)

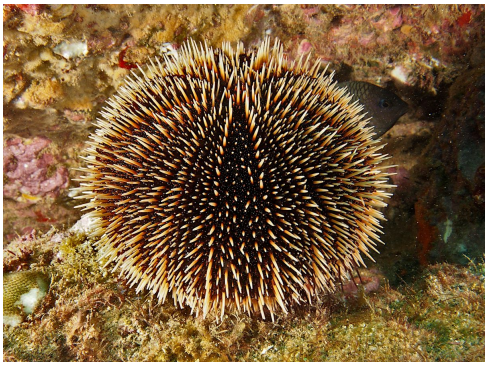


6. How can the structure of a trophic pyramid in a reef system differ from most trophic pyramids? (1 point)

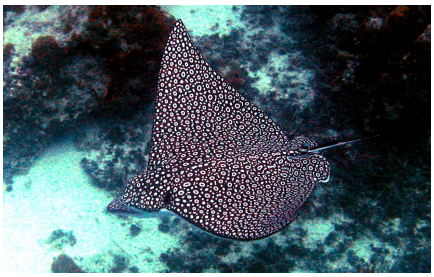
7. Give two reasons for this difference. (2 points)

8. Identify the following organisms by their common name. Note: parts c-f are on the following page (1 point each)

a. 

b. 

c. 

d. 

e. 

f. 

9. The organism in part (8b) can be an indicator of reef health. It helps corals compete for limited reef space and by controlling the growth of _____, feeding on it. (1 point)
10. On the other hand, an overabundance of this species is a detriment to (carnivorous/herbivorous) species that likewise should thrive in a healthy reef. (1 point)

11. An overabundance of the species in (8a) can be an indicator of _____. (1 point)

12. The presence of sharks also indicates that the reef is healthy as sharks are _____. (2 points)

Coral Fauna:

1. a. What does the crown of thorns starfish feed on? (1 point)

   b. How does it feed? (1 point)

2. a. What is an example of a species the triton feeds on? (1 point)

   b. How does it feed? (1 point)

3. Which of the following organisms prey on moray eels? (1 point)
   a. Grouper
   b. Parrot fish
   c. Snapper
   d.Triton

4. Most species of parrot fish are _____, meaning that they may change their sex at some point during their development. (1 point)

5. What is the difference between protandry and protogyny? (1 point)

6. True or False: Throughout its life, the humphead wrasse changes its coloration. (1 point)

7. The gorgonia forms a symbiotic relationship with what type of species? (1 point)

8. True or False: Like coral, the sea cucumber is a stationary species that exhibits little independent movement. (1 point)

9. Briefly describe the ecological niche held by sea cucumbers. (2 points)

10. Briefly explain the symbiotic relationship the banded coral shrimp holds with fish and other organisms living in the coral reef. (2 points)

11. Does the adult banded coral shrimp have any natural predators? (1 point)

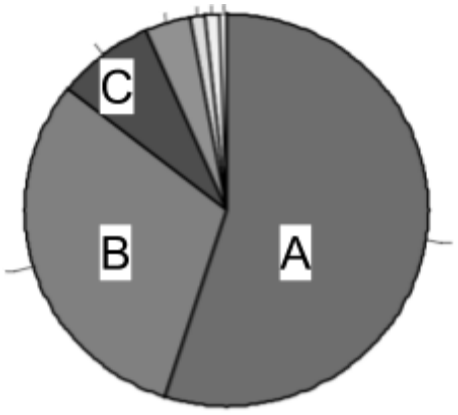
Part III: (35 points)

1. For what salinity is the maximum density of water greater than the freezing point? (1 point)

2. Does the density of water increase or decrease with an increase of salinity? (1 point)

3. 1 gram of salt (NaCl) is dissolved in 50 grams of water. What is its salinity? (2 points)

Ion	Formula
Sodium	Na^+
Magnesium	Mg^{2+}
Calcium	Ca^{2+}
Potassium	K^+
Strontium	Sr^{2+}
Chloride	Cl^-
Sulfate	SO_4^{2-}
Bicarbonate	HCO_3^-
Bromide	Br^-
Borate	$H_3BO_4^-$
Fluoride	F^-
Totals	11



Weight% of Major Ions

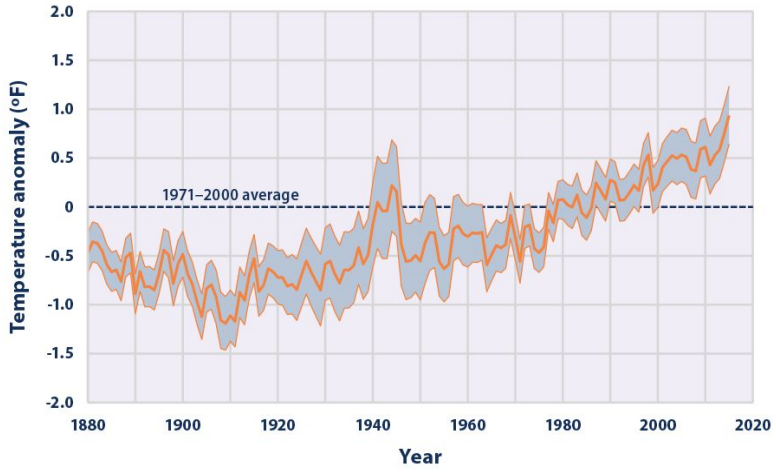
4. Above is a table of major ions found in seawater, besides a representation of their percent weight. What ions are represented by sections A, B, and C? (3 points)

5. What is the name of the device used to measure turbidity? (1 point)

6. Is the device you named in part 5 appropriate for measuring the turbidity of fast, shallow stream? If not, what device would be? If so, what device would NOT be? (2 points)

7. What are two factors that could be responsible for high turbidity in freshwater? (2 points)

8. Heat exchange with the atmosphere is largely responsible for the surface temperature of the oceans. Are the following factors responsible for heating the water deeper down *to a significant degree*? Answer yes or no for each factor. (3 points)
    a. Geothermal heating from the Earth's interior
    b. Turbulent mixing of wind and waves
    c. Thermodynamic conduction within the water

9. 

[1]

The above graph depicts the average global sea surface temperature from 1880 to 2015. The shading in the graph represents the uncertainty in the data. What is a possible cause for the trend depicted? (3 points)

10. Which of the following is an illness a human can contract after interaction with water with a high fecal coliform count? (1 point)
    a. tetanus
    b. typhoid fever
    c. malaria
    d. tuberculosis

11. Would you expect dissolved oxygen levels in a body of water to increase or decrease overnight? What biological process is the most significant contributor this increase or decrease? (2 points)

12. A local factory funnels hot water into a stream. How do you expect the resulting change in temperature to impact the dissolved oxygen level of the steam? (1 point)

13. Define Biological Oxygen Demand and describe how it is measured. (2 points)

14. Vaterite, _____, and _____ are the three most common calcium carbonate crystal structures. Circle the form that is lead soluble in water. (2 points)

15. Which biological structures may be constructed from aragonite or calcite? (2 points, circle all that apply)

[1] Source: https://www.epa.gov/climate-indicators/climate-change-indicators-sea-surface-temperature

a. Shells
b. Hair
c. Nails
d. Skeletons

16. Describe situations in which it may be important to monitor total solids in a body of water. (3 points)

17. How would an increase in total solids impact the fitness of organisms adapted to low temperatures in a body of water and why? (2 points)

18. What is the major advantage of nitrate isotopic analysis? (2 points)

Part IV (13 points)
Find the salinity of the given solution to within 1%. You may calibrate your salinometer with the 5% solution. (13 points)